

claude-windows / a9fd4426-f234-4e4a-bfb4-1f2c8255c661

Metadata

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- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-muneem\ a9fd4426-f234-4e4a-bfb4-1f2c8255c661\subagents\workflows\wf_076beba8-59c\agent-a747560b6d4ffc21a.jsonl`  
- SHA-256: `bb1b223ee5cc53fe85869b4f65e75d83aac0cbe30dcf1a620cbeb077d6573c5d`  
- Source modified: `2026-05-31T04:17:58+00:00`  
- Imported at: `2026-07-05T16:48:20+00:00`  
- project: `wf_076beba8-59c`  
- session_id: `a9fd4426-f234-4e4a-bfb4-1f2c8255c661`
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Transcript

1. user (2026-05-31T04:16:19.484Z)

Web Searcher: Borderless-table accuracy & financial-number hallucination/verification (contrarian/skeptical)

Research question: "Identify the best LOCAL (self-hosted, single NVIDIA CUDA GPU) vision / document-AI / OCR models ? and, as a clearly-flagged approval-gated fallback only, paid cloud API services ? for parsing BANK STATEMENT PDFs into structured transaction tables (date, narration/description, withdrawal/debit, deposit/credit, running balance), for a LOCAL-FIRST personal-finance tool ("muneem").

Target documents: Indian bank statements (HDFC, Axis, ICICI, AU Small Finance) ? dense, multi-column, often SEMI-RULED or borderless tables, sometimes multiple accounts per statement; both born-digital (text layer present) and the harder cases where pdfplumber's extract_tables/coordinate clustering is unreliable (e.g. Axis, whose column layout varies between statements).

HARD CONSTRAINTS (a model that violates these is unusable for us):

1. MUST run 100% LOCALLY for statement contents (highly sensitive). NO cloud APIs by default. Paid/cloud is acceptable ONLY as an explicitly-approved fallback ? still report the best cloud options but label them clearly as "cloud/sensitive ? approval-gated, not default".
2. MUST be usable from Python (transformers / vLLM / Ollama / ONNX Runtime / native lib).
3. License MUST be permissive & commercially safe for BOTH the CODE and the MODEL WEIGHTS: MIT, BSD, Apache-2.0, ISC, HPND. AVOID and explicitly FLAG anything GPL/LGPL/AGPL, CC-BY-NC / non-commercial, research-only, or custom "community"/acceptable-use licenses with restrictions. This is critical ? many document-AI models and their training datasets are non-commercial (e.g. LayoutLMv3 weights are CC-BY-NC; Surya and Marker use GPL-or-commercial dual licensing; several VLM weights ship custom restrictive licenses). Verify the WEIGHTS license, not just the repo license.
4. Hardware: a single consumer/prosumer NVIDIA GPU. Give VRAM tiers (~8-12 GB consumer, ~16-24 GB) and which models/quantizations fit each.

Evaluate and compare these LOCAL approaches ? for each give: code license, MODEL-WEIGHTS license (state explicitly), latest release / maintenance status as of 2025-2026, VRAM need + quantization options, Python integration path, and reliability specifically on DENSE FINANCIAL TABLES / multi-column statements:

- Open vision-language models (VLMs) for document/table understanding: Qwen2-VL and Qwen2.5-VL (note per-size license ? which sizes are Apache-2.0 vs restricted), MiniCPM-V 2.6, InternVL2 / 2.5, Microsoft Phi-3.5-vision, Mistral Pixtral, Meta Llama-3.2-Vision (flag the Llama Community License restrictions), Moondream2, GOT-OCR2.0, olmOCR and other OCR-specialized VLMs.
- OCR-specialist / document-AI pipelines: PaddleOCR + PP-Structure (table recognition), docTR, EasyOCR, Tesseract (already used), Microsoft Table Transformer / TATR (check code AND PubTables-1M dataset/weights license), Donut, Surya (FLAG license), Marker (FLAG license), MinerU / PDF-Extract-Kit, RapidOCR.
- Table-structure recognition specifically (cell/grid reconstruction for borderless tables): TATR, PP-Structure, UniTable, etc.

Also assess: for SEMI-RULED / borderless bank tables, is a prompt-driven VLM (page image -> JSON

transactions) more reliable than a classic OCR + table-structure pipeline? What are the accuracy/hallucination risks of VLMs on financial NUMBERS (transposed digits, dropped/duplicated rows), and how should output be VERIFIED? (muneem already machine-checks parses by balance reconciliation ? opening + sum(deposits) - sum(withdrawals) == closing, plus per-row running-balance walks and cross-checks against the bank's printed summary ? so any model's output can be automatically validated, which should inform the recommendation.)

Finally, briefly cover the best PAID/CLOUD fallbacks (clearly flagged approval-gated for sensitive data): AWS Textract (AnalyzeExpense / Tables), Google Document AI, Azure AI Document Intelligence, Reducto, LlamaParse, Anthropic/OpenAI vision ? note any self-hosting option and data-handling terms.

Deliver a RANKED shortlist of the most promising LOCAL options for muneem (NVIDIA GPU, Python, commercial-license-safe), each with license (code + weights), VRAM tier, maturity, strengths/weaknesses for dense bank-statement tables, and a concrete recommendation: which local model(s) to prototype for muneem's statement parsing and how they'd combine with the existing deterministic parsers + balance-reconciliation verification (e.g. VLM as a fallback/cross-check when deterministic parsing fails to reconcile). Cite sources and call out any license uncertainty explicitly."

Your angle: ****Borderless-table accuracy & financial-number hallucination/verification (contrarian/skeptical)**** ? Directly tests the core methodological question ? are VLMs more reliable than OCR+structure pipelines on semi-ruled tables, and what are their numeric-hallucination failure modes ? to justify muneem's balance-reconciliation-as-validator design. Search query: `VLM OCR hallucination financial numbers transposed digits dropped rows accuracy benchmark bank statement borderless table extraction vs deterministic parser reconciliation`

Task

Use WebSearch with the query above (or a refined version). Return the top 4-6 most relevant results.

Rank by relevance to the ORIGINAL question, not just the search query. Skip obvious SEO spam/content farms.

Include a short snippet capturing why each result is relevant.

Structured output only.

2. assistant (2026-05-31T04:16:22.579Z)

I'll research this question about VLM/OCR hallucination on financial numbers and the borderless-table accuracy debate. Let me load the WebSearch tool first.

3. assistant (2026-05-31T04:16:26.261Z)

I have the search tools loaded. Let me run several targeted searches in parallel to cover the contrarian/skeptical angle on VLM hallucination of financial numbers and the OCR-vs-VLM debate for borderless tables.

4. user (2026-05-31T04:16:35.380Z)

Web search results for query: "VLM OCR hallucination financial numbers transposed digits dropped rows accuracy benchmark bank statement table extraction"

Links: [{"title": "The Ultimate Guide to 99.9% Accurate Bank Statement Extraction: OCR + AI for Any Format | Statement Extract Blog | Statement Extract", "url": "https://statementextract.com/blogs/ultimate-guide-accurate-bank-statement-extraction-ocr-ai/"}, {"title": "VLM Benchmarks Explained", "url": "https://datature.io/glossary/vlm-benchmarks"}, {"title": "OCR vs VLM-OCR: Accuracy Benchmark for Scanned Documents in 2025", "url": "https://www.dataunboxed.io/blog/ocr-vs-vlm-ocr-naive-benchmarking-accuracy-for-scanned-documents"}, {"title": "FinCriticalED: A Visual Benchmark for Financial Fact-Level OCR", "url": "https://arxiv.org/pdf/2511.14998"}, {"title": "Mistral OCR vs. Gemini Flash 2.0: Comparing VLM OCR Accuracy | Reducto", "url": "https://reducto.ai/blog/vlm-ocr-accuracy-mistral-gemini"}, {"title": "Bank Statement OCR Data Extraction : a Complete Guide", "url": "https://www.koncile.ai/en/ressources/extract-data-from-bank-statements-with-"}]

```
ocr"},"title":"FAITH: A Framework for Assessing Intrinsic Tabular Hallucinations in Finance","url":"https://arxiv.org/pdf/2508.05201"}]]
```

I found relevant research and information about VLM OCR hallucination and financial data extraction accuracy. Here are the key findings:

VLM Hallucination Issues in Financial OCR

LLMs may misread numbers (guessing a blurry "6" as an "8") and can fabricate plausible data when unsure about numbers or dates, filling gaps with estimated information. The danger of financial hallucination is significant because LLMs are trained to sound plausible, making this "a compliance catastrophe for finance".

Benchmark Research

The FinCriticalED benchmark is a fact-centric visual benchmark evaluating whether OCR and vision-language systems preserve financially critical evidence, containing 859 real-world financial document pages with 9,481 expert-annotated facts spanning numeric, temporal, monetary unit, reporting entity, and financial concept types. Results reveal a clear gap between lexical accuracy and factual reliability, with numerical values and monetary units emerging as the most vulnerable fact types.

Table Extraction Challenges

Structural errors can occur where content from within tables moves outside of tables, and rows can be shifted upward incorrectly.

VLM Accuracy on Complex Documents

VLM models like Qwen and Mistral achieved text similarity scores up to 3-4 times higher than traditional OCR on complex scanned documents, making VLM-based approaches justified for high-value information like financial statements where accuracy is paramount.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

5. user (2026-05-31T04:16:37.389Z)

Web search results for query: "vision language model table extraction reliability vs OCR pipeline benchmark borderless tables PubTables OmniDocBench"

Links: [{"title":"High?accuracy OCR with reliable table extraction","url":"https://llms.reducto.ai/high-accuracy-ocr-table-extraction"}, {"title":"OCR for Tables: How to Extract Structured Data from Documents","url":"https://www.llamaindex.ai/blog/ocr-for-tables"}, {"title":"Top 7 Open Source OCR Models - KDnuggets","url":"https://www.kdnuggets.com/top-7-open-source-ocr-models"}, {"title":"7 Best Open-Source OCR Models 2025: Benchmarks & Cost Comparison | E2E Networks","url":"https://www.e2enetworks.com/blog/complete-guide-open-source-ocr-models-2025"}, {"title":"Beyond Text Extraction: The 2025 Open OCR Revolution Powered by Vision-Language Models | by TechEon | Medium","url":"https://atul4u.medium.com/beyond-text-extraction-the-2025-open-ocr-revolution-powered-by-vision-language-models-89ad33d36bbf"}, {"title":"Enhancing Table Recognition with Vision LLMs: A Benchmark and Neighbor-Guided Toolchain Reasoner","url":"https://arxiv.org/html/2412.20662v2"}, {"title":"Enhancing Table Recognition with Vision LLMs: A Benchmark and Neighbor-Guided Toolchain Reasoner","url":"https://arxiv.org/pdf/2412.20662"}, {"title":"Visual-TableQA: Open-Domain Benchmark for Reasoning over Table Images","url":"https://arxiv.org/pdf/2509.07966"}]]

Links: [{"title":"Benchmarking PDF Parsers on Table Extraction with LLM-based Semantic Evaluation","url":"https://arxiv.org/html/2603.18652v1"}, {"title":"[2512.10888] PubTables-v2: A new large-scale dataset for full-page and multi-page table extraction","url":"https://arxiv.org/abs/2512.10888"}, {"title":"PubTables-v2: A new large-scale"}]]

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dataset for full-page and multi-page table
extraction", "url": "https://arxiv.org/pdf/2512.10888"}, {"title": "OmniDocBench: Benchmarking
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Ouyang_OmniDocBench_Benchmarking_Diverse_PDF_Document_Parsing_with_Comprehensive_Annotations_CVP
R_2025_paper.pdf"}, {"title": "PubTables-1M: Towards comprehensive table extraction from
unstructured documents", "url": "https://arxiv.org/pdf/2110.00061"}, {"title": "OmniDocBench:
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_Parsing_with_Comprehensive_Annotations"}, {"title": "GitHub - opendatalab/OmniDocBench: [CVPR
2025] A Comprehensive Benchmark for Document Parsing and Evaluation ·
GitHub", "url": "https://github.com/opendatalab/OmniDocBench"}, {"title": "OmniDocBench:
Benchmarking Diverse PDF Document Parsing with Comprehensive
Annotations", "url": "https://arxiv.org/pdf/2412.07626"}, {"title": "OmniDocBench: Benchmarking
Diverse PDF Document Parsing with Comprehensive
Annotations", "url": "https://arxiv.org/html/2412.07626v1"}]
```

Based on the search results, here's what I found about vision language models for table extraction, OCR pipelines, borderless tables, and the mentioned benchmarks:

Vision Language Models vs. OCR Pipelines

VLLM-based methods have emerged to bypass traditional OCR pipelines for visual table analysis and understanding. For complex documents with tables, multi-column layouts, and mixed content, modern VLM-based models like DeepSeek-OCR, PaddleOCR-VL, and OlmOCR-2 significantly outperform Tesseract, handling layout preservation and structured output that Tesseract struggles with.

Borderless Tables

Borderless tables rely on whitespace alignment instead of visible gridlines, making positional inference more complex for conventional OCR engines. However, VLLMs are more effective on tables with unclear row borders, as they comprehend hierarchical structure and produce correct recognition results, demonstrating superior adaptability and robustness compared to traditional OCR models.

PubTables Benchmark

PubTables-v2 provides the first large-scale benchmark for full-page and multi-page table extraction (467K single pages with 548K tables and 9,172 multi-page documents). Multi-page table recognition is a key gap in current models' capabilities.

OmniDocBench

OmniDocBench contains bounding box information for tables on each PDF page along with corresponding table recognition annotations, with 1651 PDF pages covering 10 document types, 5 layout types, and 5 language types. OmniDocBench includes 142 tables with complex backgrounds, 81 containing formulas, 150 with merged cells, and 7 vertical tables.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

6. assistant (2026-05-31T04:16:41.226Z)

These are strong. The FAITH, FinCriticalED, and the "Enhancing Table Recognition with Vision LLMs" benchmark papers are directly on-point for the skeptical angle. Let me run a few more targeted searches to find the specific contrarian/practitioner perspectives on VLM numeric hallucination and the verification/reconciliation angle.

7. user (2026-05-31T04:16:51.153Z)

Web search results for query: "LLM table extraction silent failures duplicated rows skipped rows numbers wrong production document parsing reliability skeptical"

Links: [{"title": "Extract Data From PDF Tables: Row-Level Guide | LlamaIndex", "url": "https://www.llamaindex.ai/blog/extracting-repeating-entities-from-documents"}, {"title": "LLM Table Extraction: Own the Infrastructure or Offload It | Logic", "url": "https://logic.inc/resources/llm-table-extraction"}, {"title": "Handling LLM Challenges: Retry Logic, Error Recovery, and Reliability | TableFlow", "url": "https://tableflow.com/blog/handling-llm-challenges"}, {"title": "LLM APIs Aren't Complete Document Parsers: Why | LlamaIndex", "url": "https://www.llamaindex.ai/blog/llm-apis-are-not-complete-document-parsers"}, {"title": "Why PDF to Markdown Fails for LLM-Based Document ...", "url": "https://unstract.com/blog/why-pdf-to-markdown-ocr-fails-for-ai-document-processing/"}, {"title": "Why table structure extraction fails: A deep dive into real-world challenges", "url": "https://www.upstage.ai/blog/en/why-table-structure-extraction-fails-a-deep-dive-into-real-world-challenges"}, {"title": "Benchmarking PDF Parsers on Table Extraction with LLM-based Semantic Evaluation", "url": "https://arxiv.org/html/2603.18652v1"}, {"title": "Benchmarking PDF Parsers on Table Extraction with LLM-based Semantic Evaluation", "url": "https://arxiv.org/pdf/2603.18652"}, {"title": "PromptPort: A Reliability Layer for Cross-Model Structured Extraction", "url": "https://arxiv.org/pdf/2601.06151"}]

Based on my search results, I found comprehensive information about LLM table extraction challenges in production environments. Here are the key findings:

Silent Failures and Data Corruption

LLM-based processing introduces a dangerous failure mode called silent corruption, where LLMs may hallucinate content, omit sections, or truncate output without raising explicit errors, and because the resulting output still "looks valid," these failures often go unnoticed until extraction results are wrong.

Reliability Issues in Production

LLM systems are non-deterministic in ways traditional infrastructure isn't, and table extraction compounds that: models update, document formats drift, and edge cases in merged cells or multi-page tables surface after launch, not before. Additionally, merged cells, nested headers, and tables spanning multiple pages cause hallucinations even in the best available systems.

Duplicated and Skipped Rows

LLMs bring flexibility in extraction but that flexibility comes with some failure modes, including dealing with long lists of repeating entities. Mid-tier parsers such as GPT-4 Mini and Gemini 2.5 Flash produce broad distributions of partial errors rather than clean successes or outright failures.

Output Format Issues

Structured extraction with LLMs fails in production not because models lack understanding, but because output formatting is unreliable across models and prompts? a prompt that returns clean JSON on GPT-4 may produce fenced, prose-wrapped, or malformed output on other models.

Information Loss During Parsing

Markdown fundamentally discards critical layout and metadata essential for accurate extraction; bounding boxes, reading order, confidence scores, and complex table structures are often lost, and this information loss can silently corrupt data and make downstream AI processing unreliable.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

8. user (2026-05-31T04:16:51.665Z)

Web search results for query: "why we don't use LLMs for invoice bank statement number

extraction hallucination deterministic parser validation checksum"

Links: [{"title": "Understanding LLM AI Hallucinations in Data Extraction Models | Veryfi", "url": "https://www.veryfi.com/data/ai-hallucinations/"}, {"title": "LLMs vs OCR Data Extraction: Which One Should You Use?", "url": "https://www.klippa.com/en/blog/information/llms-vs-ocr-software/"}, {"title": "Building Reliable Invoice Extraction Prompts That Handle Edge Cases | Thomas Wiegold Blog", "url": "https://thomas-wiegold.com/blog/building-reliable-invoice-extraction-prompts/"}, {"title": "A 2026 Guide to AI Invoice Data Extraction | Best Invoice OCR", "url": "https://unstract.com/blog/ai-invoice-processing-and-data-extraction/"}, {"title": "LLMs for Structured Data Extraction from PDFs in 2026", "url": "https://unstract.com/blog/comparing-approaches-for-using-llms-for-structured-data-extraction-from-pdfs/"}, {"title": "Why Large Language Models May Fall Short for Precise Data Extraction in Banking !: By Prasoon Mukherjee", "url": "https://www.finextra.com/blogposting/24436/large-language-models-are-not-a-solution-for-precise-data-extraction-in-banking"}, {"title": "Invoice Extraction Using LLMs: How It Works and What to Expect | Invoice Data Extraction", "url": "https://invoicedataextraction.com/blog/invoice-extraction-using-llm"}, {"title": "Automating Bank Statement Analysis with LLMs & RAG Techniques", "url": "https://www.c-sharpcorner.com/article/automating-bank-statement-analysis-with-llms-rag-techniques/"}, {"title": "FAITH: A Framework for Assessing Intrinsic Tabular Hallucinations in Finance", "url": "https://arxiv.org/pdf/2508.05201"}, {"title": "Towards reducing hallucination in extracting information from financial reports using Large Language Models", "url": "https://arxiv.org/pdf/2310.10760"}]

Based on the search results, here's a comprehensive overview of why LLMs are not ideal for invoice and bank statement number extraction:

Hallucination and Unreliability

LLMs have a problematic tendency to be confidently wrong, and if you ask for a value, they'll give you one?even if it means hallucinating. LLMs may introduce errors or hallucinate incorrect data, which refers to instances where the model generates incorrect or completely fabricated information that was not present in the input data.

Lack of Purpose-Built Design

These models are not purpose-built for the precision and reliability that accounting requires, and a primary concern is the lack of guaranteed accuracy and consistency. LLMs might generate plausible or probable information that doesn't actually exist in the documents, which can lead to inaccurate data extraction that is critical in banking where precision is essential for financial calculations.

Consistency Issues

It is common for an LLM to struggle to return consistent outputs, and you might have to continuously tweak the prompt or specify expected output formats. General models like OpenAI GPT-4 can hallucinate or misinterpret data, inventing figures or transposing numbers, which can lead to critical errors in your financial records.

Need for Validation

Since LLMs can hallucinate or extract the wrong data, we need to ensure that there are automated evaluations in place so that we can trust the data that the LLM is responding with. Some solutions attempt to mitigate this through deterministic validation approaches, such as dual-LLM approaches that ensure extracted data is either correct or flagged as NULL, preventing hallucinations or errors that can occur with single-LLM systems.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

9. assistant (2026-05-31T04:16:56.578Z)

Excellent results. The Finextra piece ("Why Large Language Models May Fall Short for Precise Data Extraction in Banking") is exactly the contrarian/skeptical practitioner angle. The FAITH arXiv paper appears repeatedly as the canonical financial-tabular-hallucination benchmark. Let me verify the two most load-bearing arXiv papers (FAITH and the Vision LLM table benchmark) to confirm their findings and licensing relevance, and confirm the Finextra/Logic/dataunboxed angles are substantive.

10. user (2026-05-31T04:17:00.722Z)

The server returned HTTP 403 Forbidden.

The response body was not retrieved. If this URL requires authentication, use an authenticated tool (e.g. `gh` for GitHub, or an MCP-provided fetch tool) instead of WebFetch.

11. user (2026-05-31T04:17:04.295Z)

FAITH: Assessing Tabular Hallucinations in Finance LLMs

Based on the PDF metadata, **FAITH** is a framework designed to evaluate how large language models generate false information when processing financial tables—a critical concern for financial NLP applications.

Key Failure Modes

The research identifies specific ways models mishandle financial data:

- **Number transposition**: Models swap numerical values incorrectly
- **Row-level errors**: Systems drop or duplicate entire rows from tables
- **Extraction failures**: Models struggle to accurately extract numbers from structured financial data

Core Conclusion

The framework emphasizes that financial number extraction from tables poses serious reliability challenges. The authors highlight that models cannot be trusted to autonomously handle financial data without verification mechanisms. Their work underscores the necessity for "Model verification and validation" in financial contexts, indicating that human review or systematic verification processes remain essential before deploying LLM outputs in financial decision-making.

The research was presented at the 6th ACM International Conference on AI in Finance (ICAIF '25), suggesting its relevance to both academic and industry stakeholders concerned with responsible AI deployment in high-stakes financial domains.

[Binary content (application/pdf, 1006.3KB) also saved to C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-muneem\9fd4426-f234-4e4a-bfb4-1f2c8255c661\tool-results\webfetch-1780201019681-c2bd5q.pdf]

12. assistant (2026-05-31T04:17:09.525Z)

The FAITH paper is confirmed as directly on-point (note: it's an LLM-on-text-tables benchmark, ICAIF '25 ? the hallucination findings generalize, but it's not specifically VLM-on-images, which I'll flag). The Finextra blog is 403-blocked, but its title and search snippet are clear enough on relevance. Let me verify the Vision-LLM table recognition benchmark paper, which is the most directly relevant VLM-vs-OCR comparison for the borderless-table question.

13. user (2026-05-31T04:17:19.671Z)

Vision LLMs for Table Recognition: Benchmark Summary

Overview

This paper introduces a comprehensive benchmark for evaluating Vision Large Language Models (VLLMs) on table recognition tasks across hierarchical dimensions (cell, row, column, and table-level), using three datasets: SciTSR, PubTabNet, and WTW.

Performance on Borderless/Semi-Ruled Tables

Interestingly, VLLMs show surprising resilience with missing borders. The analysis reveals that "modifications such as fading or removing these borders have minimal impact on the performance of VLLMs in TR tasks." The models don't heavily rely on structural borders for recognition, instead depending on cell content positioning.

However, **VLLMs underperform traditional OCR methods on the WTW dataset** (wild scenarios with complex table variations). While traditional methods achieved 93.86 TEDS-Struct, VLLMs with NGTR only reached 52.03 points, indicating persistent challenges with irregular table structures.

VLLM Failure Modes

Key failure patterns identified:

1. **Row-Column Bias**: VLLMs prefer column-structured information over rows, struggling more with "row-related tasks" due to higher row similarity.
2. **Geometric Sensitivity**: Performance drops substantially under geometric deformations?tilting at 40 degrees causes severe accuracy loss.
3. **Complex Structures**: The models "tend to ignore cells lacking semantic content," causing failures on tables with many empty cells and dense packing.
4. **Context Confusion**: VLLMs can be "misled by its own reasoning...when encountering atypical table structures."

Neighbor-Guided Toolchain Reasoner (NGTR)

NGTR is a training-free framework addressing image quality bottlenecks through:

- **Toolkit**: Five lightweight tools (upscaling, border enhancement, noise reduction, binarization, detection/cropping)
- **Neighbor Retrieval**: Uses similarity-based retrieval to find comparable training examples
- **Tool Selection**: Leverages neighbor experiences to generate multiple tool invocation plans
- **Reflection Module**: Iteratively validates each preprocessing step to prevent information loss

Results show significant improvements: +10.57 TEDS points on PubTabNet and +12.02 on WTW, demonstrating that guided preprocessing substantially enhances VLLM capabilities for challenging inputs.

14. assistant (2026-05-31T04:17:24.910Z)

This is a superb result for the contrarian angle. It provides empirical nuance: VLLMs are resilient to missing borders (supports the VLM-for-borderless hypothesis) BUT underperform traditional OCR on "wild" complex tables (52 vs 94 TEDS-Struct), and have specific failure modes (row bias, ignoring empty cells, geometric sensitivity). That directly tests muneem's core methodological question.

I now have strong coverage across the spectrum: financial-hallucination benchmarks (FAITH, FinCriticalED), the VLM-vs-OCR table benchmark, production silent-failure practitioner posts (Logic, LlamaIndex), and the banking-skeptic angle (Finextra). I have enough to deliver a ranked shortlist.

15. user (2026-05-31T04:17:49.157Z)

Structured output provided successfully

16. assistant (2026-05-31T04:17:58.719Z)

I've returned the ranked shortlist via StructuredOutput. Summary of the contrarian/skeptical findings for muneem:

The evidence directly validates muneem's balance-reconciliation-as-validator design. Three high-relevance findings, each flagged with caveats per attribution discipline:

1. **FAITH** (arXiv 2508.05201, ICAIF '25) ? canonical benchmark for the exact failure modes (transposed numbers, dropped/duplicated rows); concludes models can't be trusted without verification. Caveat: tests text-table LLMs, not VLMs on page images.
2. **Vision-LLM table benchmark** (arXiv 2412.20662) ? the sharpest answer to the core methodological question, cutting both ways: VLLMs are resilient to missing borders (good for semi-ruled tables) but score 52 vs 94 TEDS-Struct against traditional methods on "wild" complex tables, with documented row-bias and empty-cell-skipping failures.
3. **FinCriticalED** (arXiv 2511.14998) ? visual fact-level benchmark showing numeric values and monetary units are the most vulnerable fact types, and that text-similarity accuracy ? factual correctness. This is the key warning that justifies validating via balance arithmetic, not text-overlap metrics.

Two flags worth noting: the Finextra banking-skeptic source returned HTTP 403, so its entry rests on the search snippet rather than full-text verification. And the strongest VLM benchmarks here (2412.20662, 2511.14998) are evaluated on scientific/financial-report tables and "wild" web tables, not Indian bank statements specifically ? the failure-mode patterns transfer, but no source benchmarks HDFC/Axis/ICICI layouts directly.